A Two-Level Capacity Configuration Method for Hybrid Energy Storage System in Renewable Energy Base

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***Abstract*—This paper proposes a planning-operation integrated optimization method for hybrid energy storage system (HESS) capacity configuration in renewable energy bases. A two-level optimization model is established, where the upper level determines system investment, while the lower level simulates operation of systems integrated with battery energy storage (BES) and compressed air energy storage (CAES). Case study results demonstrate that the optimal hybrid energy storage configuration is approximately BES = 14% × 2h, CAES = 10% × 10h of renewable energy installed capacity. Multi-dimensional evaluation shows that hybrid energy storage effectively improves renewable energy utilization rate by 2.8% and reduces CO_2 emissions by 4.4% compared to single-storage configurations, but with a slight increase of levelized cost of energy (LCOE). The proposed method provides a comprehensive framework for hybrid energy storage system planning, offering insights for the configuration of hybrid energy storage system.**

***Index Terms*—Hybrid energy storage, renewable energy base, capacity optimization, battery energy storage, compressed air energy storage**

I. Introduction

Driven by the "carbon peak" and "carbon neutrality" goals, the development of renewable energy has been accelerated. The rapid development of renewable energy technologies has led to the emergence of large-scale renewable energy bases, which integrate wind power, photovoltaic generation, thermal power and energy storage systems for long-distance transmission. However, the inherent intermittency and volatility of renewable energy sources pose significant challenges to power system stability and renewable energy utilization efficiency.

As the proportion of renewable energy increases, the difficulty of daily, medium-term and long-term balance of renewable energy bases will increase. A single energy storage technology is difficult to simultaneously meet the power and electricity balance requirements of different time scales [1]. For example, the most mature electrochemical energy storage (lithium-ion batteries, flow batteries, sodium-ion batteries) can basically meet the regulation requirements of short-time scales (hours), but the system regulation of long-time scales (days or weeks) cannot be met.

Hybrid Energy Storage System (HESS) combines two or more storage technologies with complementary characteristics to address the multi-timescale challenges effectively, including application requirements of power fluctuations over days, weeks and even seasonal periods [2]. Compared with single battery energy storage, hybrid systems can leverage the advantages of each technology while compensating the limited capability of single storage [3]. This approach enables potentially more efficient and cost-effective solutions for renewable energy integration compared to single-technology storage systems.

At present, the development of hybrid energy storage technology is still in its early stages because of the complexity of the energy storage system itself and the significant differences in the technical maturity of various energy storage systems. Researchers have conducted extensive studies on the configuration of hybrid energy storage systems, which is the prerequisite for the operation and control of hybrid energy storage systems. Ref. [4] proposed a frequency division algorithm to optimize the configuration of hybrid energy storage systems on the industrial load side, Similar statistical methods have also been proposed in other research papers, such as Fourier transform [5], and other data processing methods [6].

A lifecycle optimization method is proposed for the configuration of hybrid energy storage systems by using a mathematical programming method, and the results show that the hybrid energy storage system can effectively reduce the cost of renewable energy systems [7]. Other optimization-based methods have also been proposed in other research papers, but are solved by heuristic methods, such as particle

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swarm optimization [8], genetic algorithm [9], sparrow search algorithm [10] and other multi-objective optimization methods [11]. In this paper, we use the idea based on mathematical optimization but implement it through a two-level optimization method.

This paper focuses on the optimal configuration of hybrid energy storage systems in the planning stage for renewable energy bases. Our main contributions are:

(a) Modeling of multiple resources including HESS: We develop comprehensive mathematical models for different energy storage technologies, including lithium-ion battery energy storage (BES) and compressed air energy storage (CAES) systems. These models capture the operational characteristics and technical constraints of each storage technology, providing the foundation for capacity optimization.

(b) A two-level capacity configuration framework: We formulate a two-level optimization framework that integrates planning decisions (storage capacity and power ratings) with operational strategies (charging/discharging schedules). The upper level optimizes the investment decisions considering capital costs and operational benefits, while the lower level determines optimal operational strategies under typical scenarios. This integrated approach ensures that planning decisions are validated through operational simulation.

The remainder of this paper is organized as follows: Section II presents the modeling framework for hybrid energy storage systems. Section III describes the case study and analysis of results. Section IV concludes the paper with key findings and future research directions.

II. Modeling and Problem Formulation

A. System Overview

Renewable energy bases primarily rely on wind and photovoltaic power generation, supported by thermal power units as peak-shaving resources, and are equipped with new-type energy storage systems for integrated wind-solar-thermal-storage bundled transmission. This paper considers hybrid energy storage as the energy storage configuration form for renewable energy bases, used to smooth renewable energy output fluctuations and energy transfer, with the regulation space determined by the difference between renewable energy output curves and transmission load demand curves. The system structure is shown in Fig. 1.

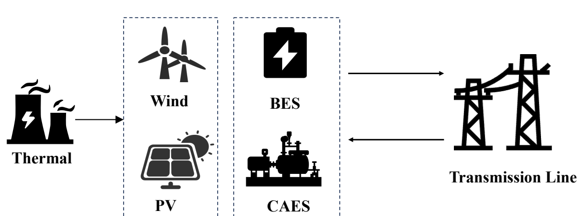

Fig. 1. Illustration of the renewable energy base generation scenario

B. Two-Level Optimization Configuration Model

This paper fixes the installed capacity of wind power, photovoltaic, and thermal power units for convenience. The capacity of electrochemical storage and compressed air energy storage is determined through the optimization configuration model.

To obtain the optimal capacity configuration for hybrid energy storage, this paper adopts a general planning-operation integrated method, establishing a two-level optimization model. The upper level (investment level) determines the capacity of hybrid energy storage resources with the objective of maximizing total system investment returns, while the lower level (operation level) simulates and optimizes renewable energy base operations with the objective of maximizing operational benefits.

The investment level problem is designed as a deterministic optimization problem to obtain the hybrid energy storage configuration under minimum investment cost. The operation level considers cost-benefit under multiple uncertainty factors, determined by the coordinated operation of renewable energy output, hybrid energy storage systems, and thermal power units, which falls under stochastic programming problems in optimization. Through clustering methods to establish typical scenario sets, stochastic programming problems can be transformed into deterministic optimization problems.

The upper-level investment planning model can be expressed as problem 1.

$$\max_{x^U} \quad C^S(x^U) = -C^E(x^U) + \mathbb{E}[C^R(x^U, x^D; \xi)] \tag{1a}$$

$$\text{s.t.} \quad C^E(x^U) = \frac{1}{365}\left[\sum_{b\in\{\text{BES,CAES}\}} k_b\left(C_b^p P_b^R + C_b^e e_b^R\right)\right] \tag{1b}$$

$$k = \frac{r(1+r)^{l_\tau - 1}}{(1+r)^{l_\tau - 1} - 1}, \quad \tau \in \{\text{BES}, \text{CAES}\} \tag{1c}$$

$$\gamma_b^{\min} P_b^R \le e_b^R \le \gamma_b^{\max} P_b^R \tag{1d}$$

where x^U, x^D are the decision variables of the upper and lower levels respectively, $C^S(x^U)$ is the total system investment return, $C^E(x^U)$ is the system investment cost, $C^R(x^U, x^D)$ is the system operation return, and k is the equivalent annual value coefficient. ξ represents the uncertainty variables in system operation, referring to wind and photovoltaic output uncertainty. C^p, C^e are the power cost coefficient and energy capacity coefficient of the energy storage system respectively. k is the equivalent annual value coefficient, and is calculated by the equipment life l_τ and the interest rate r. P_b^R, e_b^R are the rated power and rated energy capacity of the energy storage system respectively, and $\gamma_b^{\min}, \gamma_b^{\max}$ are the minimum and maximum energy storage energy-to-power ratios.

The lower-level optimization operation model can be expressed as problem 2.

$$\min_{x^D} \quad \mathbb{E}[C^R(\overline{x^U}, x^D; \xi)] = \sum_{w \in \Omega} \rho_w C^R_w(\overline{x^U}, x^D_w) \tag{2a}$$

$$\text{s.t.} \quad C^R(\overline{x^U}, x^D; \xi) = R_{\text{elec}} - C_{\text{g}} - C_{\text{curt}} - C_{\text{shed}} - C_{\text{BES}} - C_{\text{CAES}} \tag{2b}$$

$$R_{\text{elec}} = \sum_{t=1}^{T} r^e P^{\text{dc}}_t \tag{2c}$$

$$C_{\text{g}} = \sum_{t=1}^{T} \left[a(P_{g,t})^2 + bP_{g,t} + c + d \cdot \text{SU}_t\right] \tag{2d}$$

$$C_{\text{curt}} = \sum_{t=1}^{T} \alpha \left(P^{\text{curt}}_{\text{pv},t} + P^{\text{curt}}_{\text{w},t}\right) \tag{2e}$$

$$C_{\text{shed}} = \sum_{t=1}^{T} \beta \left(P^{\text{load}}_t - P^{\text{dc}}_t\right) \tag{2f}$$

$$C^{\text{om}}_b = k^{\text{om}}_b \cdot IC_b, \quad b \in \{\text{BES}, \text{CAES}\} \tag{2g}$$

The additional constraints are:

$$\sum_{t=1}^{T} P_{RE,t} \geq \theta \sum_{t=1}^{T} P^{\text{total}}_{RE,t}, \quad RE \in \{\text{PV}, \text{W}\} \tag{2h}$$

$$\text{Operation constraints of power sources and storages} \tag{2i}$$

where Ω, ρ_w are the scenario set and scenario probability respectively. The stochastic programming problem is simplified to a deterministic programming problem (2a) through scenario clustering, and the wind and solar power uncertainties ξ are considered as different scenarios Ω with probability ρ_w. The system operation return includes electricity sales revenue R_{elec}, thermal power unit operation cost C_{g}, renewable energy curtailment loss C_{curt}, load shedding penalty C_{shed}, and energy storage operation and maintenance cost C^{om}_b. In addition, the lower-level operation needs to ensure a minimum renewable energy utilization rate θ and operation constraints for various power sources and storages.

C. Power Source Constraints

(1) Thermal power unit operation constraints can be expressed as equation 3.

$$\lambda_g x_{g,t} P^R_g \leq P_{g,t} \leq x_{g,t} P^R_g \tag{3a}$$

$$-r_g P^R_g \leq P_{g,t} - P_{g,t-1} \leq r_g P^R_g \tag{3b}$$

$$\sum_{i=t_0}^{t} x_{g,i} \geq T_U(x_{g,t} - x_{g,t-1}) \tag{3c}$$

$$\sum_{i=t_1}^{t} (1 - x_{g,i}) \geq T_D(x_{g,t-1} - x_{g,t}) \tag{3d}$$

$$\text{SU}_t \geq x_{g,t} - x_{g,t-1}, \quad \text{SU}_t \geq 0 \tag{3e}$$

$$\sum_{t=1}^{T} \text{SU}_t \leq 1 \tag{3f}$$

where $x_{g,t}$ is the on-off status of the thermal power unit, P^R_g is the rated power of the thermal power unit, r_g is the ramp rate limit of the thermal power unit, T_U is the minimum start-up time, T_D is the minimum shutdown time, SU_t is the start-up cost auxiliary variable, and λ_g is the power range coefficient of the thermal power unit. These include unit power range (3a), ramp rate limits (3b), minimum start-up time (3c), minimum shutdown time (3d), start-up cost auxiliary variables (3e), and maximum start-stop times of 1 (3f).

(2) Renewable energy (wind power, photovoltaic) operation constraints can be expressed as equation 4.

$$P_{RE,t} = P^{\text{raw}}_{RE,t} - P^{\text{curt}}_{RE,t}, \quad RE \in \{\text{PV}, \text{W}\} \tag{4a}$$

$$0 \leq P^{\text{curt}}_{RE,t} \leq P^{\text{raw}}_{RE,t}, \quad RE \in \{\text{PV}, \text{W}\} \tag{4b}$$

where $P^{\text{raw}}_{RE,t}$ is the raw renewable energy output, $P^{\text{curt}}_{RE,t}$ is the curtailed renewable energy output, and $P_{RE,t}$ is the real renewable energy output. (4a) and (4b) are the mathematical constraints of them.

D. Hybrid Energy Storage System

The hybrid energy storage system includes electrochemical storage and compressed air energy storage, with operation constraints summarized as equation 5.

$$SOC_{b,t+\Delta T} = SOC_{b,t} + \eta^{ch}_b P^{ch}_{b,t} \Delta T - \frac{1}{\eta^{dis}_b} P^{dis}_{b,t} \Delta T, \quad b \in \{\text{BES}, \text{CAES}\} \tag{5a}$$

$$0 \leq P^{ch/dis}_{b,t} \leq x^{ch/dis}_{b,t} P^R_b, \quad b \in \{\text{BES}\} \tag{5b}$$

$$\alpha x^{dis}_{c,t} P^R_c \leq P^{dis}_{c,t} \leq x^{dis}_{c,t} P^R_c, \quad c \in \{\text{CAES}\} \tag{5c}$$

$$\beta x^{ch}_{c,t} P^R_c \leq P^{ch}_{c,t} \leq x^{ch}_{c,t} P^R_c, \quad c \in \{\text{CAES}\} \tag{5d}$$

$$x^{ch}_{b,t} + x^{dis}_{b,t} \leq 1, \quad b \in \{\text{BES}, \text{CAES}\} \tag{5e}$$

$$0 \leq SOC^b_t \leq e^R, \quad b \in \{\text{BES}, \text{CAES}\} \tag{5f}$$

$$SOC^b_0 = SOC^b_T = 0.5 e^R_b, \quad b \in \{\text{BES}, \text{CAES}\} \tag{5g}$$

$$P^{out}_{b,t} = P^{dis}_{b,t} - P^{ch}_{b,t}, \quad b \in \{\text{BES}, \text{CAES}\} \tag{5h}$$

where (5a)(5f)(5g)(5h) are the typical charge/discharge output and SOC status, and (5e) is the charge-discharge exclusive constraint. Meanwhile, (5b) is the power range constraint of BES, (5c) is the discharging power range constraint of CAES, (5d) is the charging power range constraint of CAES. Since charge and discharge process is not symmetric, the power range constraint of CAES is different from that of BES. The HESS model can be flexibly extended to other energy storage technologies because it describe only the power and SOC status, but more detailed constraints should be added such as (5b)(5c)(5d) in this case composed of BES and CAES.

E. Other Constraints

- System energy balance:

$$P_{load,t} = P_{g,t} + P_{pv,t} + P_{w,t} + P^{out}_{b,t} + P^{out}_{c,t} \tag{6}$$

- DC load power balance, where ΔP^L_t represents the power supply gap:

$$P^{\text{dc}}_t + \Delta P^L_t = P^{\text{load}}_t \tag{7}$$

III. Case Study

A. System Setup and Parameters

The renewable energy base supporting power source installed capacity for wind power, photovoltaic, and thermal power are configured as 400MW, 1000MW, and 400MW respectively. According to the adopted renewable energy output data, the design utilization hours for wind power and photovoltaic are 2680h and 1755h respectively.

The renewable energy power source investment and operation parameters are shown in Table I, where the installed costs for wind power and photovoltaic are 5000 CNY/kW and 4000 CNY/kW respectively, with annual operation and maintenance ratios of 3%. The thermal power unit operation parameters are designed as follows: the minimum start-up time is 6 hours, minimum shutdown time is 4 hours, and the generation cost coefficients a, b, c are 0.01 CNY/MW^2, 300 CNY/MW, and 20 CNY respectively. In addition, the penalty coefficients in the optimization problem are $\alpha = 300$ CNY/MWh and $\beta = 5000$ CNY/MWh, representing renewable energy curtailment loss and load shedding penalty coefficients.

TABLE I
Renewable Energy Power Source Investment and Operation Parameters

Power Source Type	Wind Power	Photovoltaic	Unit
Installed Cost	5000	4000	CNY/kW
Design Utilization Hours	2680	1755	h
Annual O&M Cost	3%	3%	-

B. Evaluation Index

The evaluation system for configuration optimization includes three dimensions: power indicators, economic indicators, and environmental indicators, used for multi-dimensional evaluation of the rationality of energy storage capacity configuration.

1) Operation Indicator Evaluation: The operation indicator evaluation content includes:

1) Load power supply reliability:

$$R_{load} = \frac{\sum_t P_{load,t} - P_{dc,t}}{\sum_t P_{load,t}} \times 100\% \tag{8}$$

2) Renewable energy curtailment rate:

$$\theta = \frac{\sum_t P^{\text{cur}}_{pv,t} + P^{\text{cur}}_{w,t}}{\sum_t P^{\text{total}}_{pv,t} + P^{\text{total}}_{w,t}} \times 100\% \tag{9}$$

3) Energy storage system utilization rate:

$$U_b = \frac{\sum_t P^{dis}_{b,t}}{e^R_b} \times 100\%, \quad b \in \{\text{BES}, \text{CAES}\} \tag{10}$$

2) Economic Evaluation: Levelized Cost of Energy (LCOE) is used to describe the full lifecycle generation cost of planned projects, calculated as the ratio of annualized investment and operation costs to renewable energy generation, specifically:

$$\text{LCOE} = \frac{(AC^{\text{inv}} + C^{\text{om}})(P^{\text{rate}}_{pv}, P^{\text{rate}}_{w}, P^R_b, P^R_c, e^R_b, e^R_c)}{\sum_t^T P_{pv,t} + P_{w,t}} \tag{11}$$

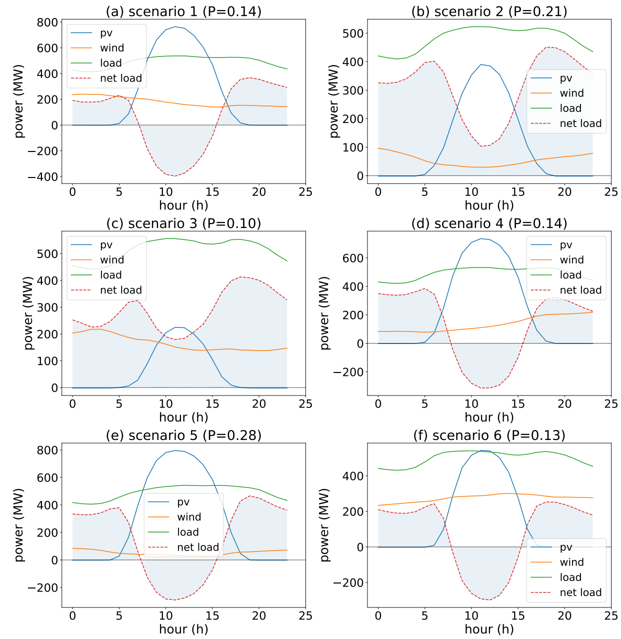

Fig. 2. Scenario clustering results

3) Environmental Benefit Assessment: System carbon emission assessment:

$$E_{\text{CO}_2} = \alpha_{\text{coal}} \cdot P_{\text{gen}} \tag{12}$$

where α_{coal} is the carbon emission coefficient of coal-fired power generation, calculated based on standard coal consumption of 300g/kWh, with coal-fired unit carbon emission intensity of approximately 850g CO_2/kWh.

C. Clustering Method and Scenarios

The typical K-means clustering method is used for scenario generation. As shown in Fig. 2, the scenarios are clustered into $K = 6$ clusters, the probability of each typical scenario is calculated as $\rho_w = [0.14, 0.21, 0.10, 0.14, 0.28, 0.13]$.

D. Optimal Configuration and Results

A simple search method is used to find the optimal comfiguration of hybrid energy storage. The initial configurations are searched under different energy storage configurations, with fixed energy storage durations for electrochemical storage and compressed air energy storage of 2h and 10h respectively. The parameter search space is $e^R_b \in [0, 2000]$, $e^R_c \in [0, 2000]$, with current installed capacity, and the search space is sufficiently large. The results are shown in Fig. 3.

The optimal configuration for the hybrid energy storage system is BES = 200MW/400MWh, CAES = 140MW/1400MWh, with optimal objective function value of 476,775. Under the condition that compressed air energy storage technology is not available, the optimal configuration for electrochemical storage is BES = 400MW/800MWh, with optimal objective function value of 447,316. Under the condition that electrochemical storage technology is not available, the optimal configuration for

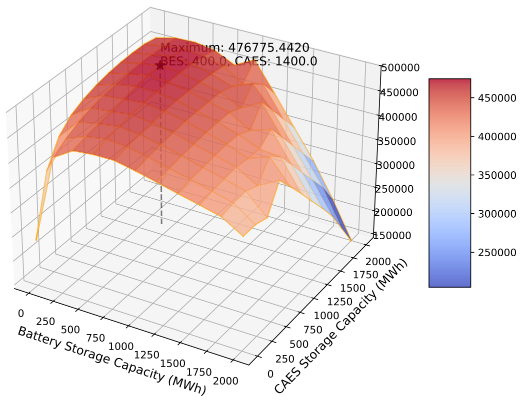

Fig. 3. Objective function value under different configurations

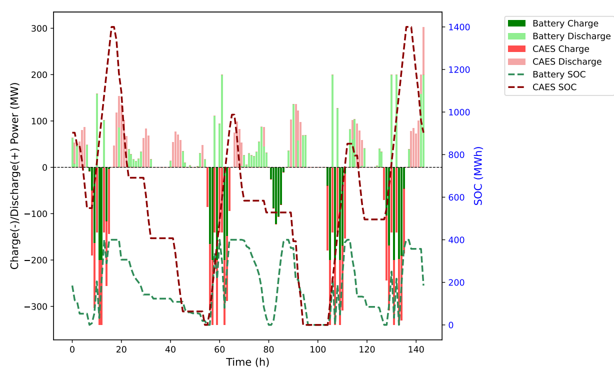

Fig. 4. SOC curves of optimal hybrid energy storage configuration

compressed air energy storage is CAES = 160MW/1600MWh, with optimal objective function value of 462,839.

The results show that the hybrid energy storage configuration effectively improves renewable energy utilization rate, reduces curtailment rate and carbon emission levels, but also brings certain LCOE increase. The optimal configuration for hybrid energy storage is approximately BES = 14% × 2h, CAES = 10% × 10h of renewable energy installed capacity.

The SOC curves and output results of the optimal hybrid energy storage configuration are shown in Fig. 4 and Fig. 5. It shows that the hybrid energy storage configuration can effectively smooth the output fluctuations of renewable energy, and the output of thermal power units is relatively stable.

E. Configuration Results Comparison

For the configuration results comparison, the following three cases are selected:

- Case 1: Optimal configuration with only battery energy storage (BES = 400MW/800MWh, CAES = 0);
- Case 2: Optimal configuration with only compressed air energy storage (BES = 0, CAES = 160MW/1600MWh);
- Case 3: Optimal configuration with hybrid energy storage (BES = 200MW/400MWh, CAES = 140MW/1400MWh);

The generation ratio of different configurations is shown in Table II. It shows the hybrid energy storage configuration (Case 3) achieves the best performance of 89.8% in renewable energy utilization rate and 60.52% in renewable energy generation with comparison to the other two configurations. The results show that the hybrid energy storage configuration can effectively improve the utilization rate of renewable energy.

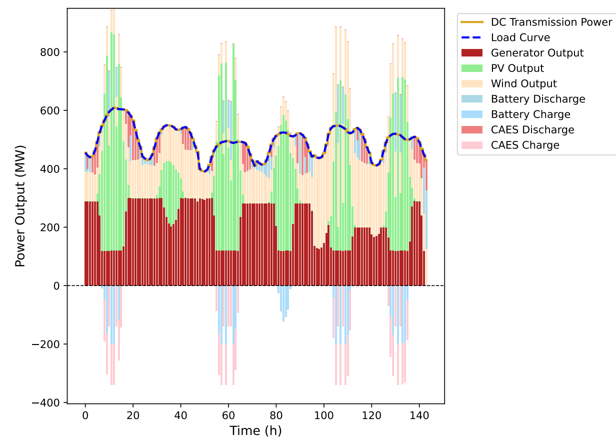

Fig. 5. Output results of optimal hybrid energy storage configuration

The comparison of comprehensive indicators is shown in Table III. The results show that the hybrid energy storage configuration does not improve the utilization rate of storages, either for battery energy storage (1.59 cycle/day) or compressed air energy storage (0.56 cycle/day), and the LCOE is slightly higher than the other two configurations (0.347 CNY/kWh). However, the hybrid energy storage configuration can decrease the CO_2 emission level, and promote the carbon emission reduction of the system.

TABLE II
GENERATION RATIO UNDER DIFFERENT CONFIGURATIONS

Generation ratio(%)	Case 1	Case 2	Case 3
Load shortage	0.00	0.00	0.00
PV curtailment	23.20	17.78	16.82
Wind curtailment	5.40	6.63	1.20
Renewable energy utilization	840	87.00	**89.80**
PV generation	29.69	31.78	32.16
Wind generation	27.16	26.81	28.37
Thermal power generation	436	463	42.67
Renewable energy generation	56.85	58.59	**60.52**

TABLE III
COMPARISON OF COMPREHENSIVE INDICATORS UNDER DIFFERENT CONFIGURATIONS

Indicator	Case 1	Case 2	Case 3
BES investment (10^4 CNY)	9.60	0.00	80
CAES investment (10^4 CNY)	0.00	12.80	11.20
Storage investment (10^4 CNY)	9.60	12.80	16.00
Annual income (10^4 CNY/year)	7.10	7.28	7.83
BES utilization rate (cycle/day)	**1.59**	0.00	**1.59**
CAES utilization rate (cycle/day)	0.00	**0.61**	0.56
Cost recovery period (year)	10	10	10
Investment return rate (%)	10.2	10.0	10.3
LCOE (CNY/kWh)	0.345	0.340	**0.347**
Annual CO_2 emission (10^4 ton/year)	163.99	165.00	**157.73**

IV. CONCLUSIONS

This chapter proposes a planning-operation integrated hybrid energy storage capacity optimization configuration method for the renewable energy base of wind-solar-thermal-storage integrated transmission scenario. The two-level optimization model is first established, and then the search strategy is adopted to find the optimal configuration. The case study results show that the optimal configuration for hybrid energy storage is approximately BES = 14% × 2h, CAES = 10% × 10h of renewable energy installed capacity. By comparing the comprehensive indicators of different configurations, the improvement in renewable energy utilization rate and reduction in carbon emission level of hybrid energy storage configuration is verified. The proposed method provides a comprehensive framework for hybrid energy storage system planning, offering insights for the configuration of hybrid energy storage system.

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